

# George Gu

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## EDUCATION

### University of Michigan

*Bachelor of Science in Engineering, Computer Science*

May 2028

Ann Arbor, MI

- **GPA:** 3.81/4.00
- **Coursework:** Machine Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Theory of Computation, Computer Organization, Web Systems, Computer Security

## EXPERIENCE

### Capital One

*Technology Intern*

June – August 2026

Richmond, VA

- Architected data pipelines via scheduled AWS Lambda workers to ingest daily snapshots from internal asset-management systems, yielding clean tabular datasets to train predictive models on application lifecycle and business criticality.
- Engineered Python REST APIs orchestrating data-attestation and asset lifecycle compliance workflows, securing them with parameterized PostgreSQL queries and role-based access control across multi-level corporate domains and org hierarchies.
- Authored 20+ automated integration tests covering mock database transactions, cross-region failures, and network-timeout edge cases, lifting repository-wide coverage to 91% (100% on all new files) for the compliance service.

### BoilerVault

*Founding Engineer (Contract)*

January 2026 – Present

West Lafayette, IN

- Architected a multi-tenant operations platform with FastAPI, PostgreSQL, and Next.js/TypeScript for a 3-campus student storage business, implementing JWT auth, role-based access control, and 20+ REST endpoints validated by 140+ tests.
- Engineered idempotent webhook pipelines across 4 Stripe accounts and 3 WordPress booking sites, reconciling out-of-order payment events to automate customer, order, and billing creation with 0 manual entry across \$130K+ in gross revenue.
- Replaced a failing Zapier/spreadsheet workflow by migrating 2,000+ legacy records via three-tier matching (exact email, rapidfuzz fuzzy name, Stripe charge fallback) and deploying on Railway/Vercel as the business's production system of record.

### Nexteer Automotive

*Software Engineering Intern*

May – August 2025

Saginaw, MI

- Built an LLM-powered IDE extension in Python and TypeScript that parsed 300+ internal engineering guidelines and automated compliance checks across C and H files, integrating Azure AI and Copilot APIs into the developer workflow.
- Engineered prompt pipelines and few-shot strategies to extract and interpret rules from unstructured documentation, tuning the system to 95% violation-detection accuracy and saving engineers an estimated 8 hours of review per week.
- Deployed the tool globally across 26 sites and integrated it with internal dev pipelines, cutting manual code-review time by 87.5% and enabling reliable, real-time background screening for safety-critical embedded steering systems.

## PROJECTS

### Image Classification & Transfer Learning

*Machine Learning Developer*

January – April 2026

Ann Arbor, MI

- Implemented a Vision Transformer from scratch in PyTorch, building patch-to-token projection, a learnable classification token, sinusoidal positional embeddings, and multi-head self-attention encoder blocks for binary image classification.
- Engineered a 2-stage pretrain/fine-tune training pipeline with a 4-level data augmentation sweep, AdamW weight decay, label smoothing, and AUROC-based early stopping, reaching 0.93 validation accuracy and 0.97 validation AUROC.
- Evaluated transfer learning from an 8-class CNN across 0-3 frozen convolutional layers, diagnosing overfitting via train/validation curve divergence and confusion-matrix error analysis to guide augmentation and regularization choices.

### Scalable Search Engine

*Systems Developer*

January – April 2026

Ann Arbor, MI

- Built a multi-stage MapReduce pipeline in Python to construct an inverted index over 3,000+ documents, computing TF-IDF scores and document-normalization factors across a parallel, multi-job processing architecture for scalable retrieval.
- Implemented a REST API index server in Flask serving 3 partitioned inverted index segments, integrating PageRank-weighted cosine similarity scoring to balance TF-IDF with link authority when ranking query results by relevance.

### Resume Screener

*Machine Learning Developer*

January – May 2025

Ann Arbor, MI

- Built an NLP pipeline over a labeled dataset of 1,000+ resumes and job postings, applying tokenization, TF-IDF vectorization, and feature engineering to convert unstructured text into structured features for resume-to-posting matching.
- Designed scikit-learn and TensorFlow models to rank resumes against job postings, using sentence-transformer embeddings and RAG retrieval over job descriptions to surface a user's best-matching role, deployed via Streamlit to 100+ users.

## SKILLS/ADDITIONAL

**Languages:** Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML, CSS

**Frameworks/Developer Tools:** PyTorch, TensorFlow, Git, Docker, Kubernetes, Unix, Node.js, React.js, Flask, PostgreSQL

**Interests:** Philosophy and Metaphysics, Game Design, Bouldering, Skateboarding, Golf, Guitar, Calisthenics

## PUBLICATION

**Gu G, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, "A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle", 2023, *Antibiotics*, 12, 1509.**

doi: <https://doi.org/10.3390/antibiotics12101509>